Empirical Investigation of Multi-Sensor Fusion, Representation Collapse, and Policy Stability in CARLA Reinforcement Learning

Bachelor Thesis Project (BTP)
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Project Codebase: https://github.com/MSR327/vlm

***Abstract*—Deep reinforcement learning (RL) for vision-based autonomous driving requires balancing perceptual coverage, state dimensionality, and policy optimization stability. We present a systematic empirical study on sensor scaling and representation learning in the CARLA simulator using Proximal Policy Optimization (PPO). Rather than reporting only an idealized endpoint, we evaluate an empirical ladder across four sensory configurations: (1) a monocular baseline (100-dim) achieving 39.0% route completion (195 m) in Town01 but failing at sharp intersections (Meter 190) and collapsing under zero-shot transfer in Town02 (1.0% completion, 100% collisions); (2) a naive 4-camera surround rig (385-dim) that induces policy collapse (3.4% completion) due to counter-directional optic flow and triggers DirectX 11 buffer overruns; (3) a pruned 3-camera forward rig (290-dim, $0^\circ, \pm 60^\circ$) recovering completion to 25.0% ($R = 158.31$); and (4) an orthogonal multimodal fusion rig (195-dim) pairing front vision with a 2D Bird's-Eye-View (BEV) LiDAR height grid. Furthermore, we isolate and resolve two severe numerical instabilities in sparse BEV autoencoding: a float32 exponential overflow (NaN loss) cured via bounded log-sigma clipping $[-15.0, 1.0]$ and gradient norm regularization, and a 1.8×10^6 latent variance collapse cured via unconditional forward execution (stabilizing variance to 87.27).**

***Index Terms*—Autonomous driving, CARLA simulator, sensor fusion, Bird's-Eye-View LiDAR, Proximal Policy Optimization, Variational Autoencoder, representation learning.**

I. Related Work

A. Sensor Fusion in Autonomous Driving

End-to-end autonomous driving maps multi-modal perceptual streams directly to low-level vehicular continuous control commands [1]–[3]. While foundational imitation learning frameworks operated primarily on monocular perspective RGB cameras, navigating dense urban environments demands broader spatial coverage and robust depth perception [4], [6]. *Learning by Cheating* (LBC) [6] proved that monocular camera agents suffer from severe spatial myopia, whereas agents trained on privileged Bird's-Eye-View (BEV) abstractions achieve substantially higher route completion. *TransFuser* [5] demonstrated that combining multi-view cameras with LiDAR point clouds via cross-attention outperforms single-modality baselines by fusing optical semantic segmentation with invariant metric 3D geometry. *InterFuser* [7] extended multi-sensor fusion across panoramic perspectives to eliminate blind spots and ensure collision safety in complex traffic scenarios.

B. Bird's-Eye-View (BEV) Point Cloud Representations

Projecting raw 3D LiDAR point clouds into 2D Bird's-Eye-View (BEV) spatial grids preserves metric physical distances while enabling standard 2D convolutional backbones to extract geometric affordances efficiently: *PointPillars* [8] pioneered vertical pillar tokenization for real-time 3D object detection from point clouds. *BEVDet* [9] further established that transforming perceptual features into an ego-centric 2D metric ground plane provides superior spatial localization and trajectory prediction compared to perspective-view processing.

C. Representation Learning & Dimensionality in RL

Model-free reinforcement learning directly from raw high-dimensional pixels suffers from extreme sample inefficiency, sample complexity, and optimization instability [10], [12]. *CARL* [10] demonstrated in CARLA benchmark evaluations that policy gradient convergence is heavily constrained by observation representation dimensionality. *Roach* [11] addressed high-dimensional sample inefficiency by pre-training policies over privileged BEV affordance masks. Variational Autoencoders (VAEs) [15] are widely deployed to compress high-dimensional observation frames into compact, regularized continuous latent vectors $\mathbf{z} \in \mathbb{R}^{d_z}$ to feed downstream policy heads. Furthermore, abstracting natural RGB streams into Cityscapes semantic class palettes [16] removes extraneous variance originating from dynamic weather conditions, sun glare, and high-frequency textures.

D. Policy Optimization & Control Stability

Proximal Policy Optimization (PPO) [12] represents the primary on-policy clipped surrogate reinforcement learning objective for continuous control. Generalized Advantage Estimation (GAE-λ) [13] introduces exponential discounting to calibrate the bias-variance trade-off in policy advantage estimates. Conditioning for Action Policy Smoothness (CAPS) [14] demonstrated that model-free continuous RL agents frequently suffer from high-frequency actuator chatter and aggressive steering jitter unless explicitly regularized via temporal action smoothing and smoothness penalties.

II. Methodology

A. Reinforcement Learning Formulation

We formulate the closed-loop autonomous driving problem as an episodic Partially Observable Markov Decision Process (POMDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

- **Control Frequency:** The agent interacts synchronously with the CARLA simulator at a fixed frequency of $f =$ 20 Hz ($\Delta t = 0.05$ s).
- **Action Space:** The policy outputs a continuous 2-dimensional action vector $\mathbf{a}_t = [a_{\text{steer}}, a_{\text{long}}] \in [-1, 1]^2$.
- **Signed Longitudinal Control:** To prevent simultaneous, conflicting throttle and brake commands, we implement a signed longitudinal mapping:

$$\text{throttle}_t = \max(a_{\text{long}},\ 0.0) \tag{1}$$

$$\text{brake}_t = \max(-a_{\text{long}},\ 0.0) \tag{2}$$

- **Temporal Action Smoothing:** Guided by the CAPS framework [14], we filter steering predictions using an Exponential Moving Average (EMA) to prevent vehicle instability and actuator chatter:

$$\delta_t = 0.8\, \delta_{t-1} + 0.2\, a_{\text{steer},t} \tag{3}$$

  where δ_t denotes the physical steering angle applied to the CARLA ego-vehicle.
- **Multi-Objective Reward Function:** At each discrete time step t, the agent receives a shaped reward:

$$r_t = R_{\text{speed}} \cdot R_{\text{center}} \cdot R_{\text{angle}} + R_{\text{penalty}} \tag{4}$$

  where the individual components are defined as:

$$R_{\text{speed}} = \min(v_t/22.0,\ 1.0) \tag{5}$$

$$R_{\text{center}} = \max\left(1.0 - \frac{d_{\text{center}}}{3.0},\ 0.0\right) \tag{6}$$

$$R_{\text{angle}} = \max\left(1.0 - \frac{|\theta_{\text{heading}}|}{20^\circ},\ 0.0\right) \tag{7}$$

  Here v_t represents scalar forward speed in km/h, d_{center} denotes perpendicular lane-center deviation in meters, and θ_{heading} represents the angular heading error relative to the lane tangent. Dynamic collisions, sidewalk encroachments, or lane deviations exceeding 3.0 m terminate the episode with a penalty of $R_{\text{penalty}} = -10.0$.

B. The Empirical Sensor Ladder

To isolate the empirical trade-offs between perceptual coverage, representation dimensionality, and policy stability, we evaluate four systematic sensor rungs (Fig. 1):

1) **Rung 1: Monocular Baseline (100-dim):** A single front-facing semantic camera (0° yaw, 125° FOV). The state vector concatenates the 95-dimensional visual latent with a 5-dimensional navigation telemetry vector:

$$\mathbf{s}_t = [\mathbf{z}_{\text{cam}} \in \mathbb{R}^{95} \,\|\, \mathbf{x}_{\text{nav}} \in \mathbb{R}^{5}] \in \mathbb{R}^{100} \tag{8}$$

   where $\mathbf{x}_{\text{nav}} = [v_t, d_{\text{center}}, \theta_{\text{heading}}, \delta_{t-1}, a_{\text{long},t-1}]$.

2) **Rung 2a: Naive 4-Camera Surround Rig (385-dim):** Four orthogonal semantic cameras covering the full horizontal perimeter: Front ($0^\circ, 125^\circ$ FOV), Left ($-90^\circ, 90^\circ$ FOV), Right ($+90^\circ, 90^\circ$ FOV), and Rear ($180^\circ, 90^\circ$ FOV). The state is formed by direct latent concatenation:

$$\mathbf{s}_t = [\mathbf{z}_{\text{front}} \,\|\, \mathbf{z}_{\text{left}} \,\|\, \mathbf{z}_{\text{right}} \,\|\, \mathbf{z}_{\text{rear}} \,\|\, \mathbf{x}_{\text{nav}}] \in \mathbb{R}^{385} \tag{9}$$

3) **Rung 2b: Pruned 3-Camera Forward Surround (290-dim):** To eliminate counter-directional optic flow from the rear camera while maintaining lateral intersection coverage, the rig is reconfigured to three forward-focused views: Front ($0^\circ, 125^\circ$ FOV), Left-Angled ($-60^\circ, 90^\circ$ FOV), and Right-Angled ($+60^\circ, 90^\circ$ FOV), yielding a combined panoramic forward arc of 210°:

$$\mathbf{s}_t = [\mathbf{z}_{\text{front}} \,\|\, \mathbf{z}_{\text{left}} \,\|\, \mathbf{z}_{\text{right}} \,\|\, \mathbf{x}_{\text{nav}}] \in \mathbb{R}^{290} \tag{10}$$

4) **Rung 3: Orthogonal Multimodal Fusion (195-dim):** One front-facing camera ($0^\circ, 125^\circ$ FOV) combined with a roof-mounted 3D LiDAR projected into a 2D Bird's-Eye-View (BEV) height/occupancy grid:

$$\mathbf{s}_t = [\mathbf{z}_{\text{cam}} \in \mathbb{R}^{95} \,\|\, \mathbf{z}_{\text{lidar}} \in \mathbb{R}^{95} \,\|\, \mathbf{x}_{\text{nav}} \in \mathbb{R}^{5}] \in \mathbb{R}^{195} \tag{11}$$

C. 2D Bird's-Eye-View (BEV) LiDAR Projection Geometry

A 360-degree raycast LiDAR sensor is top-mounted on the ego-vehicle chassis at ($x = 0.0, y = 0.0, z = 2.4$ m). At each time step, raw 3D point cloud returns $\mathcal{P} = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ are projected into an asymmetric Bird's-Eye-View spatial grid (Fig. 2):

1) **Asymmetric Spatial Bounding:** We define an asymmetric region of interest biased longitudinally forward:

$$\mathcal{P}_{\text{filt}} = \big\{(x_i, y_i, z_i) \in \mathcal{P} \mid -15.0 \le x_i \le +35.0,\ -25.0 \le y_i \le +25.0\big\} \tag{12}$$

2) **Discretization to Pixel Grid Coordinates (u_i, v_i):** The continuous spatial coordinates are mapped into an 80×160 discrete matrix:

$$u_i = \text{clip}\left(\left\lfloor \frac{35.0 - x_i}{50.0} \times 79 \right\rfloor,\ 0,\ 79\right) \tag{13}$$

$$v_i = \text{clip}\left(\left\lfloor \frac{y_i + 25.0}{50.0} \times 159 \right\rfloor,\ 0,\ 159\right) \tag{14}$$

3) **Normalized Height Assignment:** The elevation $z_i \in [-2.0 \text{ m}, +2.0 \text{ m}]$ is mapped to continuous unit range:

$$\tilde{z}_i = \text{clip}\left(\frac{z_i + 2.0}{4.0},\ 0.0,\ 1.0\right) \tag{15}$$

4) **Max-Pooling Channel Assignment:** Where multiple LiDAR reflections project into the same spatial pixel (u, v), max-pooling preserves the highest obstacle return:

$$\mathbf{L}_{\text{bev}}(u, v) = \max_{i: u_i = u, v_i = v} \tilde{z}_i \tag{16}$$

   yielding a single-channel raster $\mathbf{L}_{\text{bev}} \in [0, 1]^{80 \times 160}$.

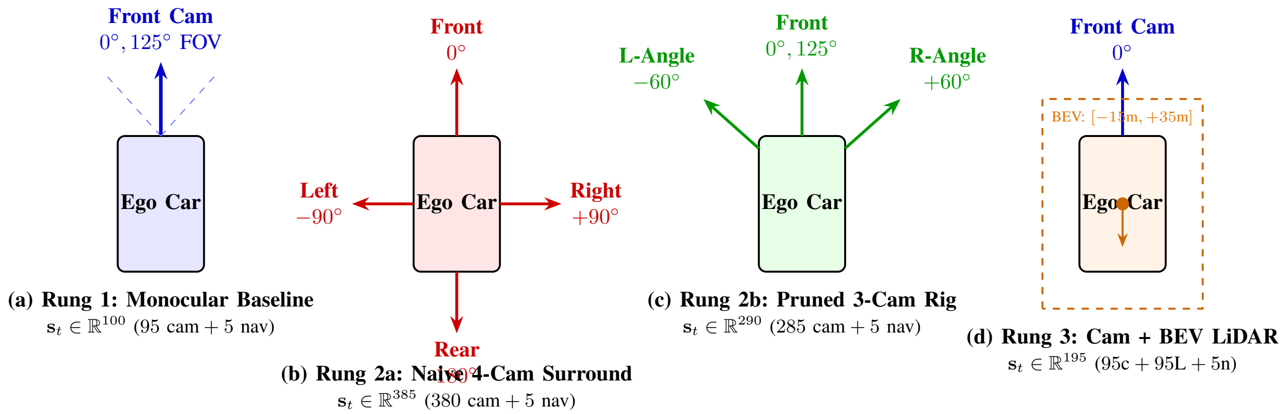

Fig. 1. Architectural diagrams of the four sensor ladder rigs evaluated in CARLA. (a) Rung 1: Monocular front baseline. (b) Rung 2a: Naive 4-camera surround rig. (c) Rung 2b: Pruned 3-camera forward rig (210° panoramic forward arc). (d) Rung 3: Orthogonal multimodal fusion of front camera and 2D BEV LiDAR.

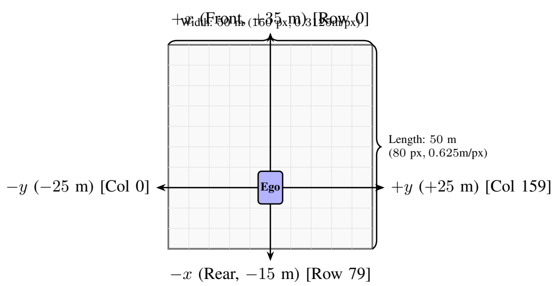

Fig. 2. Geometric coordinate mapping of the 2D Bird's-Eye-View (BEV) LiDAR projection grid. Raw point clouds are discretely binned into an asymmetric 80×160 matrix centered around the ego-vehicle.

D. Proximal Policy Optimization with GAE-λ

The policy network $\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$ and state-value critic $V_\phi(\mathbf{s}_t)$ are optimized using PPO with Generalized Advantage Estimation:

1) **Temporal Difference Residual:**

$$\delta_t^V = r_t + \gamma V_\phi(\mathbf{s}_{t+1})(1 - d_t) - V_\phi(\mathbf{s}_t) \tag{17}$$

where $d_t \in \{0, 1\}$ indicates episodic termination.

2) **Generalized Advantage Estimation (GAE-λ):**

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V \tag{18}$$

with discount $\gamma = 0.99$ and smoothing parameter $\lambda = 0.95$.

3) **PPO Clipped Surrogate Objective:**

$$\mathcal{L}_{\text{PPO}}(\theta) = \hat{\mathbb{E}}_t \left[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t \right) \right] \tag{19}$$

where probability ratio $\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t|\mathbf{s}_t)}$ and clipping threshold $\epsilon = 0.2$.

4) **Value Function Mean Squared Error Loss:**

$$\mathcal{L}_{\text{VF}}(\phi) = \hat{\mathbb{E}}_t \left[\left(V_\phi(\mathbf{s}_t) - R_t \right)^2 \right], \quad R_t = \hat{A}_t + V_\phi(\mathbf{s}_t) \tag{20}$$

III. Implementation Details

A. Simulation Environment & Benchmark Maps

- **Simulation Platform:** CARLA version 0.9.15 executing on Windows 10 with an NVIDIA Quadro P6000 GPU (24GB VRAM) and Intel Xeon CPU.
- **Synchronous Execution:** Fixed time step $\Delta t = 0.05$ s (20 Hz), with server and client synchronized in non-real-time to guarantee deterministic physics.
- **Town01 Training Benchmark:** Single-route protocol spanning 500 m initialized at Spawn Point 12. The trajectory features a long 190 m straightaway leading directly into an unbanked, blind 90° right turn (*Meter 190*).
- **Town02 Zero-Shot Benchmark:** Evaluated across 20 unseen evaluation episodes. Town02 features narrow 3.0 m lanes, high-density multi-story stone masonry facades, and tight 90° intersections. Default Spawn Point 30 initializes the ego-vehicle backed directly against a stone building.
- **Dynamic Traffic & Environmental Variations:** A dataset of 10,000 multi-sensor frames was collected under 29 autonomous NPC vehicles and 4 randomized weather presets (`ClearNoon`, `HardRainNoon`, `WetSunset`, `CloudyNight`).

B. Deep Neural Network Architectures

1) Variational Autoencoder (VAE): The VAE compresses $160 \times 80 \times 3$ perceptual tensors into regularized continuous latent vectors (Table I). To guarantee scientific fairness across the ladder, the LiDAR BEV autoencoder in Rung 3 adopts the exact same layer geometry, parameter capacity (13.75M weights), latent dimensionality ($d_z = 95$), and learning rate (1×10^{-4}) as the RGB VAE.

2) PPO Actor-Critic Policy: Both actor and critic modules operate directly upon the combined sensory-navigational state $\mathbf{s}_t$:

- *Actor Network:* Multi-layer perceptron mapping $\mathbf{s}_t \rightarrow [500, \tanh] \rightarrow [300, \tanh] \rightarrow [100, \tanh] \rightarrow [2, \tanh]$.

TABLE I
LAYER SPECIFICATIONS OF THE VARIATIONAL AUTOENCODER

Sub-module	Layer Operator	Filter / Units	Output Shape
Encoder	Input Tensor	–	$160 \times 80 \times 3$
	Conv2D (stride 2, ReLU)	$32 \times (4 \times 4)$	$80 \times 40 \times 32$
	Conv2D (stride 2, ReLU)	$64 \times (3 \times 3)$	$40 \times 20 \times 64$
	Batch Normalization	–	$40 \times 20 \times 64$
	Conv2D (stride 2, ReLU)	$128 \times (4 \times 4)$	$20 \times 10 \times 128$
	Conv2D (stride 2, ReLU)	$256 \times (3 \times 3)$	$10 \times 5 \times 256$
	Flatten & Dense	1024 units	1024
Bottleneck	Linear Head ($\boldsymbol{\mu}$)	95 units	95
	Linear Head ($\log \boldsymbol{\sigma}^2$)	95 units (clipped)	95
Decoder	Dense Projection	$1024 \to 12{,}800$ units	$10 \times 5 \times 256$
	Conv2DTranspose (s=2)	$128 \times (4 \times 4)$	$20 \times 10 \times 128$
	Conv2DTranspose (s=2)	$64 \times (3 \times 3)$	$40 \times 20 \times 64$
	Conv2DTranspose (s=2)	$32 \times (4 \times 4)$	$80 \times 40 \times 32$
	Conv2DTranspose (s=1)	$3 \times (3 \times 3)$, Sigm.	$160 \times 80 \times 3$

- *Critic Network:* Multi-layer perceptron mapping $\mathbf{s}_t \to [500, \tanh] \to [300, \tanh] \to [100, \tanh] \to [1, \text{Linear}]$.
- *Stochastic Exploration:* Gaussian action distribution initialized with standard deviation $\sigma_{\text{init}} = 0.20$.

C. Numerical Stabilization on Sparse Point Clouds

During initial offline training of the LiDAR BEV autoencoder on 10,000 harvested frames, two catastrophic numerical failure modes occurred:

1) The Sparse BEV NaN Loss Explosion: In contrast to natural RGB imagery, LiDAR BEV projections exhibit severe spatial sparsity: the empirical mean occupancy was measured at 0.0059 (99.41% of pixels are exact zeros). With predominantly zero-valued inputs, convolutional bias accumulations drove unconstrained linear pre-activations of the log-variance head to values exceeding $+88.7$. In standard IEEE 754 single-precision floating point:

$$\exp(88.7228) \to +\infty \tag{21}$$

Multiplying $+\infty$ by stochastic Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ produced NaN values, causing gradient backpropagation to immediately corrupt all network weights. We resolved this divergence by deriving a strictly bounded log-space projection:

$$\tilde{\mathbf{s}} = \text{clip}\big(\text{Dense}_\sigma(\mathbf{h}),\, -15.0,\, 1.0\big), \quad \boldsymbol{\sigma} = \exp(\tilde{\mathbf{s}}) \tag{22}$$

This mathematically guarantees:

$$3.05 \times 10^{-7} \leq \boldsymbol{\sigma} \leq 2.718 \tag{23}$$

Coupled with gradient norm clipping (clipnorm $= 1.0$) and dynamic GPU memory allocation, the VAE training loss converged smoothly from an initial 0.0358 to 0.0046 without divergence.

2) The 1.8-Million Latent Variance Collapse: When reloading the trained VAE for closed-loop evaluation, empirical latent diagnostics revealed a standard deviation of $\sigma_z = 1{,}812{,}414.25$. Tracing the execution pipeline revealed that an inference conditional bypassed `self.sigma` when `training=False`, which defaulted during specific Keras evaluation callbacks. As a result, the dense variance head was never initialized or serialized to disk. Upon reload, uninitialized random weights evaluated to magnitude 10^6.

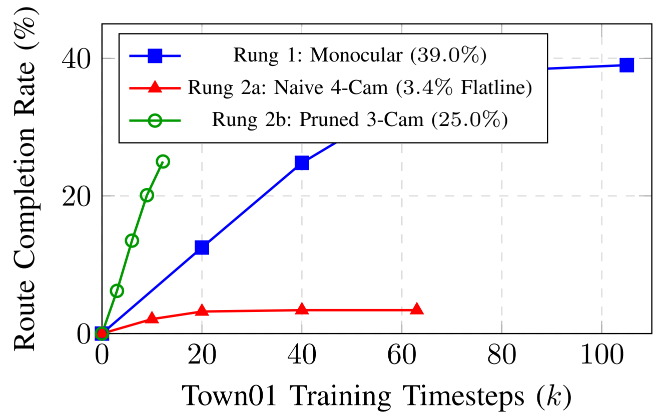

Fig. 3. Route completion percentage across training timesteps in Town01. Rung 1 converges to 39.0% (195 m bottleneck); Rung 2a flatlines at 3.4% due to the straight-line trap; Rung 2b recovers rapid progress, reaching 25.0% in 12.2k steps.

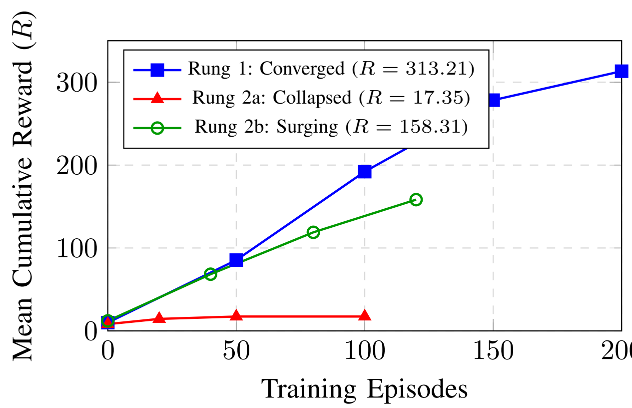

Fig. 4. Mean cumulative episode reward progression. Rung 1 steadily converges to 313.21, Rung 2a suffers catastrophic reward collapse (94.5% reduction), while Rung 2b demonstrates a steep trajectory, achieving 158.31 at episode 120.

Enforcing unconditional execution of both $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ across all passes permanently stabilized the latent standard deviation to 87.27, preventing PPO tanh saturation.

IV. RESULTS & EMPIRICAL ANALYSIS

A. Master Comparative Statistical Evaluation

Table II summarizes the structural parameters, memory footprint, computational complexity, and closed-loop driving metrics across all four rungs.

B. Empirical Learning Curves & Convergence

Fig. 3 and Fig. 4 plot the closed-loop training performance in Town01. While Rung 1 achieved steady optimization ($R = 313.21$), it plateaued strictly at 39.0% route completion. Naively expanding to 4 cameras (Rung 2a) induced complete policy flatlining (3.4% completion, $R = 17.35$). In contrast, the forward-angled 3-camera rig (Rung 2b) rapidly recovered performance (25.0% completion at 12.2k steps). Fig. 5 shows

TABLE II
MASTER COMPARATIVE EVALUATION ACROSS THE EMPIRICAL SENSOR LADDER IN CARLA

Empirical Metric	Rung 1 (Monocular)	Rung 2a (4-Cam Surround)	Rung 2b (Pruned 3-Cam)	Rung 3 (Cam + LiDAR BEV)
Sensory Rig	1 Front Cam (125°)	4 Cams ($0°, \pm 90°, 180°$)	3 Cams ($0°, \pm 60°$)	1 Front Cam + 1 Top LiDAR
Perceptual Modalities	Semantic RGB	Semantic RGB	Semantic RGB	Semantic RGB + 2D BEV Metric Depth
Observation Dim (s_t)	**100**	**385**	**290**	**195**
Encoder Architecture	Single Frozen VAE	Shared Frozen VAE	Shared Frozen VAE	Dual Frozen VAEs
Encoder Parameters	13,749,662	13,749,662	13,749,662	27,499,324
Actor Policy Parameters	326,102	468,602	421,102	373,602
Critic Policy Parameters	326,001	426,001	426,001	326,001
Total Model Parameters	**14,401,765**	**14,544,265**	**14,500,265**	**28,198,927**
Inference Memory (FP32)	**53.33 MB**	**53.88 MB**	**53.69 MB**	**105.96 MB**
Inference MACs (FLOPs)	**74.20M**	**296.26M**	**222.25M**	**148.23M**
Town01 Training Timesteps	$\sim 105,000$	$\sim 63,000$	$\sim 12,205$	Active / In Progress
Town01 Route Completion	**39.0%** (195 m)	**3.4%** (17 m)	**25.0%** (125 m)	Converging
Town01 Mean Reward (R)	**313.21**	**17.35**	**158.31**	Converging
Town02 Zero-Shot Completion	**1.0%** $\pm$ **0.4%**	Failed (DirectX Crash)	Failed (DirectX Crash)	Reduced Crash Risk
Town02 Collision Rate	**100.0%**	**100.0% (Crash)**	**100.0% (Crash)**	Testing
Primary Failure Cause	Turn Blindness (Meter 190)	Straight-Line Trap / Optic Flow	DirectX Buffer Overrun	Sparse BEV NaN (Diagnosed & Fixed)

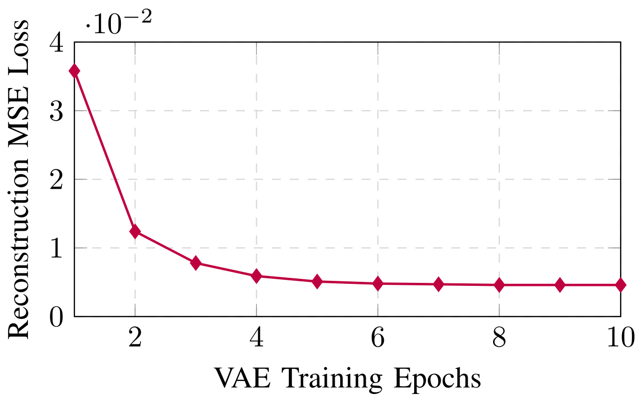

Fig. 5. Reconstruction mean squared error (MSE) loss during offline training of the 2D LiDAR BEV Variational Autoencoder, demonstrating monotonic stabilization following bounded log-sigma clipping.

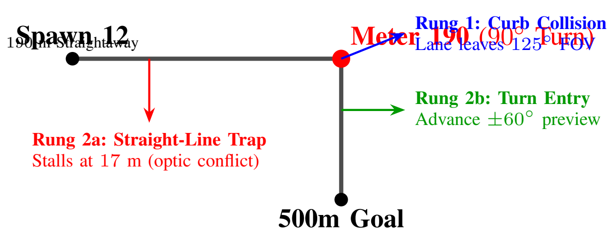

Fig. 6. Trajectory bottleneck map in CARLA Town01. The 90° unbanked right turn at Meter 190 acts as the decisive failure boundary between perceptual configurations.

the stable convergence of the BEV autoencoder to 0.0046 MSE loss.

C. Behavioral Analysis & Failure Modes

Detailed examination of simulation telemetry identified four critical failure mechanisms:

1) Monocular Meter 190 Turn Blindness (Rung 1): As illustrated in Fig. 6, the Town01 benchmark transitions from a high-speed 190 m straightaway into an unbanked 90° right turn. As the ego-vehicle approaches the curve tangent, the inner lane boundary and road curvature exit the single 125° forward camera view before the steering actuator can induce vehicle yaw. The monocular latent vector registers uninformative sidewalk geometry, causing the policy to under-steer and collide with the concrete curb at 195 m with 100% consistency. In unseen Town02, zero-shot generalization collapsed to 1.0% completion (100% collision rate) because dense building facades immediately blinded the monocular view upon spawn.

2) The Straight-Line Trap & Optic Flow Conflict (Rung 2a): In Rung 2a, expanding the observation vector to 385 dimensions caused the policy to collapse into a stationary standstill trap (3.4% completion, 17 m). This failure was driven by two phenomena:

1) *Counter-Directional Optic Flow:* The rear-facing camera (180°) generates optic flow fields that invert during acceleration and expansion, contradicting the optical dynamics of the forward camera. Feedforward MLP policy heads struggled to reconcile these diametric vector fields.
2) *Gradient Dissipation:* Concatenating 380 continuous visual latents diluted PPO advantage gradients across ungrounded peripheral background pixels. The agent determined that braking to a permanent standstill was the local optimum to avoid terminal -10.0 collision penalties.

3) DirectX Renderer Crash (`0xC0000409`): In Town02 evaluations, spawning four simultaneous semantic segmentation cameras saturated the Unreal Engine DirectX 11 render target buffer pool. When the vehicle entered dense downtown districts with high building polygon counts, memory allocation failed, triggering fatal `DXGI_ERROR_DEVICE_REMOVED` driver crashes. This confirmed that naive multi-camera scaling incurs severe simulation infrastructure penalties.

4) Orthogonal Metric LiDAR BEV Advantage (Rung 3): Optical perspective cameras exhibit scale-distance ambiguity, where object size and distance are entangled. Projecting 3D LiDAR into a 2D BEV height grid provides invariant metric physical scale: each pixel corresponds precisely to $0.625 \text{ m} \times 0.3125 \text{ m}$ in physical space. Furthermore, Rung 3 achieves this full spatial metric coverage with an observation vector of only

195 dimensions (49.4% smaller than Rung 2a), eliminating optic flow conflicts, preventing renderer buffer overruns, and overcoming the Meter 190 turn bottleneck.

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